

No.	Solution
1(a)	The total momentum of a system is constant, provided no external resultant force acts on it.
1(b)(i)	$0.160(400) - 1.20(16.0) = (0.160 + 1.20)v$ $v = 32.941 \text{ m s}^{-1}$ Change in speed of object A $= 32.941 - 400$ $= -367.06 \text{ m s}^{-1}$ $\approx -367 \text{ m s}^{-1}$
1(b)(ii)	Average force exerted on object A by object B $= \frac{\Delta p_A}{\Delta t}$ $= \frac{0.160(-367.06)}{0.450}$ $= -130.51 \text{ N}$ $\approx -131 \text{ N}$ Magnitude of average force exerted on object A by object B = 131 N
1(b)(iii)	By Newton's third law of motion, if object A exerts a force on object B, object B will exert the same type of force of equal magnitude but opposite in direction on object A. Magnitude of average force exerted on object B by object A = 131 N Direction of average force exerted on object B by object A is the same direction as the initial velocity of object A.
2(a)(i)	SI base unit of drag force, F , is kg m s^{-2}